KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
School of Computer Engineering

SAP CAPSTONE PROJECT REPORT

Order-to-Cash (O2C) Process Implementation

AuraMed Pharmaceuticals Pvt. Ltd.

Using SAP SD (Sales & Distribution) Module

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1. Executive Summary

1.1 Project Overview

This report presents the end-to-end SAP implementation of the Order-to-Cash (O2C) business process for AuraMed Pharmaceuticals Pvt. Ltd., a fictional mid-sized pharmaceutical manufacturing and distribution company headquartered in Hyderabad, India. The project is developed as a capstone submission for the B.Tech programme at KIIT University, Bhubaneswar.

The O2C cycle encompasses every business activity from the moment a customer places an order to the point at which cash is collected against the invoiced amount. In SAP, this process is primarily handled by the Sales & Distribution (SD) module, with deep integration into Financial Accounting (FI) and Materials Management (MM).

1.2 Objective of SAP Implementation

- Automate and standardise the end-to-end sales process from quotation to cash receipt.
- Eliminate manual, paper-based order management and billing workflows.
- Enable real-time credit management and customer account receivables tracking.
- Improve order fulfilment accuracy and reduce delivery lead times.
- Provide management with real-time visibility into sales performance through SAP standard reports.
- Ensure statutory compliance (GST, e-invoicing) through SAP's tax configuration framework.
- Integrate seamlessly with the Finance (FI) module for automatic revenue postings.

1.3 Selected Module — SAP SD (Sales & Distribution)

SAP SD is one of the core functional modules of SAP ERP. It manages all processes related to order fulfilment and customer relationship management from the commercial side. The module covers Pre-Sales Activities, Sales Order Management, Shipping & Delivery, Billing & Invoicing, and Credit Management. It interacts natively with SAP FI (for accounting entries), SAP MM (for inventory and goods issue), and SAP PP (for production-linked availability checks).

Note: This project covers SAP ECC 6.0 / S/4HANA SD functionality. All configuration steps reference SPRO paths applicable to S/4HANA 2021 on-premise.

2. Company Profile

2.1 Company Overview

Company Name	AuraMed Pharmaceuticals Pvt. Ltd.
Founded	2005
Headquarters	Hyderabad, Telangana, India
Industry	Pharmaceutical Manufacturing & Distribution
Turnover (FY24)	INR 850 Crore (approx.)
Employees	~4,200
ERP System	SAP S/4HANA 2021 (On-Premise)
Go-Live Date	01-April-2024 (Simulated)
Legal Structure	Private Limited Company
CIN	U24230TG2005PTC048812 (Fictional)

2.2 Business Operations

AuraMed manufactures and distributes a wide range of pharmaceutical products including prescription drugs, OTC medicines, active pharmaceutical ingredients (APIs), and nutraceuticals. The company sells to:

- Institutional clients: Government hospitals, private hospital chains (B2B)
- Wholesale distributors: Regional pharma stockists across India
- Retail pharmacy chains: Both organised and unorganised channels
- Export customers: Regulated markets (USA, UK, EU) and semi-regulated markets

2.3 Organisational Structure (Business View)

AuraMed operates through the following organisational hierarchy prior to SAP implementation:

```
  ┌──────────────────────────────────────────────┐   │         AuraMed Pharmaceuticals Pvt.
Ltd.  │  ← Corporate HQ   └──────────────────────────────────────────┘
│          ┌──────────┴──────────┐          │                    │   ┌──▼──────┐
┌──────▼──────┐   │ Domestic │          │  Exports   │  ← Business Divisions   │  Sales   │
│  Division   │   └──────────┘          └──────────┘          │                         │
┌──▼──────────┐      ┌──────▼──────────┐    │ Plants    │     │ Export Warehouse │   │ HYD
/ MUM  │      │ Mundra Port, GJ  │   └──────────┘      └──────────────────┘
```

2.4 Products Portfolio

- Prescription Drugs (Rx): Antibiotics, Cardiovascular, Oncology support lines
- OTC Medicines: Analgesics, antacids, cough & cold formulations
- APIs: Active pharmaceutical ingredients for B2B export
- Nutraceuticals: Vitamins, mineral supplements, protein powders

3. Business Requirements

3.1 Problem Statements (As-Is Situation)

Before SAP implementation, AuraMed operated primarily on a combination of legacy ERP software, MS Excel spreadsheets, and manual paper-based workflows. The following critical pain-points were identified during the Business Blueprint (BBP) phase:

Problem 1: Disconnected Order Management

Sales orders were received via email, phone calls, and fax. Order details were manually entered into multiple disconnected systems — sales team used MS Excel, the warehouse used a custom-built software, and finance used Tally. There was no single source of truth, leading to duplicate orders, data entry errors, and disputes with customers.

Problem 2: Absence of Real-Time Credit Control

There was no automated credit limit enforcement. Sales representatives would sometimes dispatch goods to customers who had already exceeded their credit limits, resulting in overdue receivables exceeding INR 45 Crore annually and write-offs due to bad debts.

Problem 3: Manual Billing and Invoice Errors

Invoices were generated manually on pre-printed stationary. GST computation, discount application, and freight charges were calculated by individual staff members using spreadsheets, leading to frequent errors, credit note issuance, and delayed payments. E-invoicing compliance with GSTN was not automated.

Problem 4: Inventory Visibility Gaps

The warehouse team had no real-time view of stock availability at the time of order booking. This led to over-promising delivery dates, short shipments, and customer dissatisfaction. Back-order management was handled entirely on phone calls.

Problem 5: Revenue Leakage Through Manual Discounting

Sales managers had no system controls over discount percentages. Field sales representatives offered ad-hoc discounts without approval workflows, eroding margins by an estimated 3–5% annually on domestic sales.

Problem 6: No MIS / Business Intelligence

Monthly sales reports were compiled by the MIS team manually from 12+ Excel files. Reports were typically delayed by 15–20 days, making it impossible for management to take timely business decisions.

3.2 Why SAP is the Recommended Solution

- SAP SD provides an integrated, end-to-end solution covering inquiry, quotation, order, delivery, billing, and payment — all in one platform.

- SAP's Credit Management module (FI-AR) enforces credit limits automatically and blocks order processing for customers who breach limits.
- Output Management in SAP SD automates invoice generation and supports GST-compliant e-invoicing via integration with the GST portal.
- Material Availability Check (ATP — Available-to-Promise) in SAP checks real-time stock at the time of order entry and gives accurate delivery dates.
- SAP's Pricing Procedure framework enforces standardised discount structures with approval workflows, preventing revenue leakage.
- SAP S/4HANA Embedded Analytics provides real-time dashboards and sales reports using SAP Fiori apps, eliminating reporting delays.

4. SAP SD Module Overview

4.1 What is SAP SD?

SAP Sales & Distribution (SD) is a core logistics module within SAP ERP that manages all business processes related to the selling, shipping, and billing of products and services. It works in close conjunction with SAP MM (inventory management), SAP FI (financial accounting), and SAP PP (production planning) to provide an integrated order fulfilment platform.

4.2 Key Components of SAP SD

Pre-Sales Activities	Manage customer inquiries and quotations (VA11 / VA21)
Sales Order Management	Core document for order processing (VA01, VA02, VA03)
Shipping & Transportation	Outbound delivery, picking, packing, goods issue (VL01N)
Billing & Invoicing	Invoice generation, credit/debit memos, proforma (VF01)
Credit Management	Credit limit checks, blocked orders management (VKM1–VKM5)
Pricing & Conditions	Flexible pricing procedure with condition types (VK11/VK12)
Master Data	Customer master, material master, customer-material info records
Output Determination	Automated printing/emailing of orders, invoices, delivery notes
Availability Check (ATP)	Real-time check of stock at order entry (MM-based)
Returns Management	Returns orders (RE), return delivery, credit memo processing

4.3 O2C Process Overview (High Level)

```
  Customer Inquiry        │        ▼   Quotation (VA21)        │        ▼   Sales Order
(VA01)  ◀──── Credit Check (FD32)        │        ▼   Delivery Order (VL01N)        │
├── Picking (LT0A / Warehouse Mgmt)        ├── Packing        └── Goods Issue (PGI)
──────────────▶ MM: Stock Reduction                                                FI:
COGS Posting        │        ▼   Billing Document (VF01) ──────────────▶ FI: AR Invoice
Posted        │        ▼   Customer Payment ──────────────────────▶ FI: Clearing (F-28)
│        ▼   Account Receivable Cleared (FB03)
```

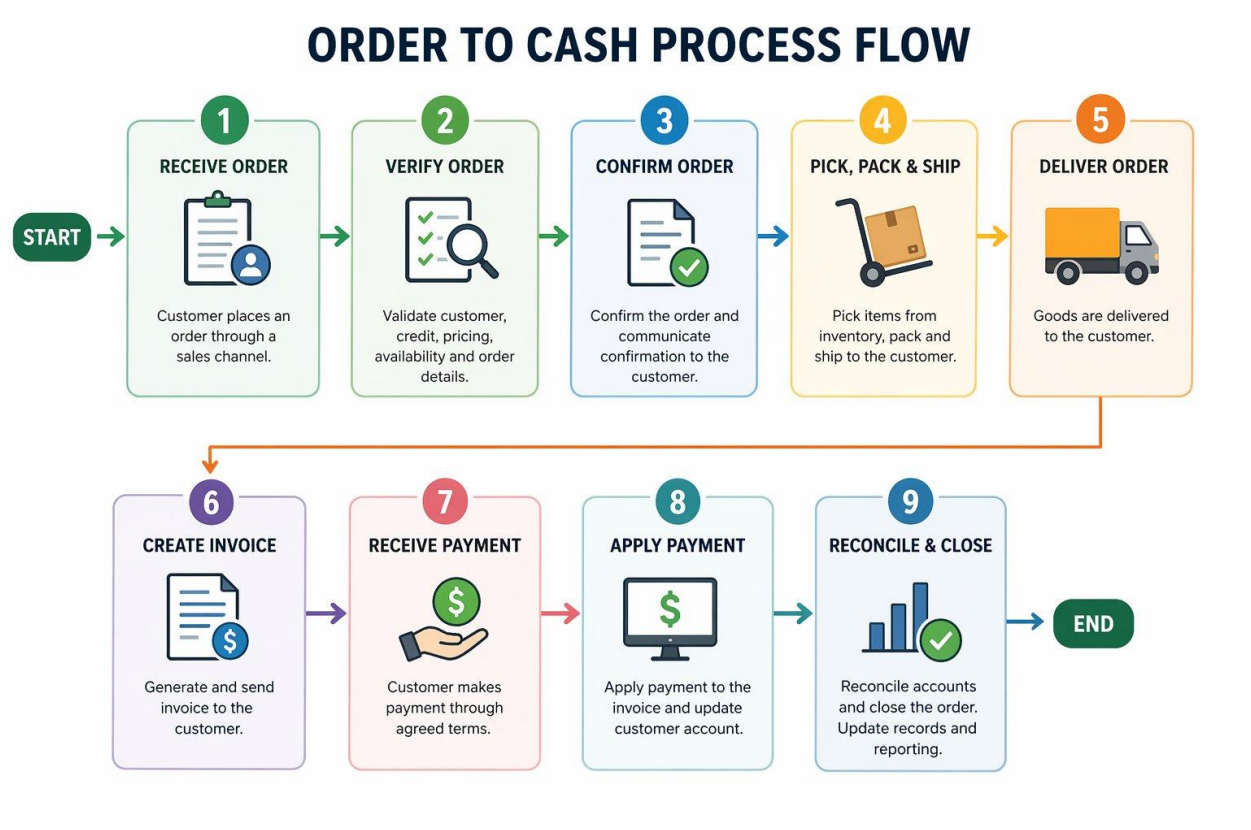

Figure 1: Order-to-Cash (O2C) Process Flow — End-to-End Overview

5. Organisational Structure in SAP

5.1 Overview

SAP's organisational structure defines the legal and logistical entities within the system. For AuraMed, the following SAP organisational units have been configured. These structures form the backbone of all SD transactions.

5.2 SAP Organisational Hierarchy for AuraMed

```
  CLIENT: 100 (AuraMed SAP System)   │   └── COMPANY CODE: AM01 (AuraMed Pharmaceuticals Pvt.
Ltd.)        │        ├── CONTROLLING AREA: AM01        │        ├── PLANT: AM10 — Hyderabad
Manufacturing Plant        │    └── STORAGE LOCATIONS:        │        ├── SL01 — Finished
Goods Warehouse        │        ├── SL02 — Raw Material Store        │        └── SL03 —
Quality Hold Area        │        ├── PLANT: AM20 — Mumbai Distribution Centre        │
└── STORAGE LOCATIONS:        │        ├── SL10 — Main Distribution Store        │
└── SL11 — Returns Store        │        ├── SALES ORGANISATION: AM_S1 — Domestic Sales India
        │    └── DISTRIBUTION CHANNEL: 10 — Institutional Sales        │    └── DISTRIBUTION
CHANNEL: 20 — Wholesale Distribution        │    └── DISTRIBUTION CHANNEL: 30 — Retail
Pharmacy        │        └── DIVISION: PH — Pharmaceuticals        │        └── DIVISION:
NU — Nutraceuticals        │        ├── SALES ORGANISATION: AM_S2 — Export Sales        │
└── DISTRIBUTION CHANNEL: 40 — Regulated Exports        │    └── DISTRIBUTION CHANNEL: 50 —
Semi-Regulated Exports        │        └── SHIPPING POINT: AM_SP1 — Hyderabad Dispatch
SHIPPING POINT: AM_SP2 — Mumbai Dispatch
```

5.3 Organisational Unit Descriptions

Client	100 — Highest level in SAP; represents the AuraMed SAP system environment.
Company Code	AM01 — Represents the legal entity 'AuraMed Pharmaceuticals Pvt. Ltd.' All FI postings happen at this level.
Plant	AM10 (Hyderabad) and AM20 (Mumbai). Plants are production/distribution units; goods movements happen here.
Storage Location	Sub-units within a plant where physical inventory is managed (Finished Goods, Raw Material, Returns).
Sales Organisation	AM_S1 (Domestic) and AM_S2 (Export). Responsible for pricing, legal obligations, and sales reporting.
Distribution Channel	Defines how products reach the customer: Institutional (10), Wholesale (20), Retail (30), Exports (40/50).
Division	Product grouping: PH (Pharmaceuticals), NU (Nutraceuticals). Used for sales analysis.
Sales Area	Combination of Sales Org + Distribution Channel + Division (e.g., AM_S1 + 10 + PH = Institutional Pharma Sales).

Shipping Point	Defines the physical dispatch location (Hyderabad warehouse or Mumbai DC) — assigned to the Plant.

6. Master Data Configuration

6.1 Customer Master

The Customer Master is the central repository of all customer-related information in SAP SD. It is created using T-Code XD01 and consists of three views:

Customer Master Structure

General Data (Client Level)	Name, address, communication details, tax classification (GSTIN), bank details. Valid for the entire SAP client.
Company Code Data	Credit control area, reconciliation account (GL), payment terms (e.g., 30 days net), dunning procedure.
Sales Area Data	Delivery plant, shipping conditions, pricing procedure assignment, output procedure, Incoterms, order currency.

Sample Customer Master — Apollo Hospital Chains Ltd.

Customer Number	100001 (internal number range)
Customer Name	Apollo Hospital Chains Ltd.
City / State	Hyderabad, Telangana
GSTIN	36AAACL1234A1ZM (Fictional)
Reconcil. Account	GL 14000001 — Trade Debtors
Payment Terms	ZT30 — Net 30 Days
Credit Limit	INR 1,50,00,000 (1.5 Crore)
Credit Control Area	AM01
Sales Area	AM_S1 / 10 / PH
Delivery Plant	AM10 — Hyderabad
Pricing Procedure	ZAMSD1 — Domestic Standard Pricing
Currency	INR — Indian Rupee
Incoterms	EXW — Ex Works, Hyderabad

6.2 Material Master

The Material Master stores all product-related data. It is created using T-Code MM01 and is organised into views. For AuraMed's pharmaceutical products, the following views are critical:

Basic Data 1 & 2	Material number, description, base unit of measure (Strip / Box / Bottle), weight, volume, material group.
Sales: Sales Org Data 1	Delivering plant, tax classification, general item category group, minimum order quantity.
Sales: Sales Org Data 2	Account assignment group (for revenue account determination), item category usage.
Sales: General / Plant Data	Availability check rule, transportation group, loading group.
MRP 1 / MRP 2	Planning type, safety stock, reorder point (used for ATP check during order entry).
Accounting 1	Valuation class, standard/moving average price — feeds into FI for COGS posting at PGI.

Sample Material Master — AMOX-500 (Amoxicillin 500mg Capsules)

Material Number	MAT-10001
Description	Amoxicillin Capsules 500mg (Strip of 10)
Industry Sector	Pharmaceutical
Material Type	FERT — Finished Product
Base Unit	STR (Strip)
Sales Unit	BOX (Box of 10 Strips)
Delivering Plant	AM10
Tax Class.	1 — Full IGST / CGST+SGST (18%)
Valuation Class	7920 — Finished Goods Pharma
Standard Price	INR 85.00 / Strip
Availability Check	02 — Individual Requirements

6.3 Customer-Material Info Record

T-Code: VD51. This master data object maps the customer's own product description/number to AuraMed's internal material number. This is essential in the pharmaceutical B2B space where hospital procurement systems use their own item codes.

Customer	100001 — Apollo Hospital Chains
AuraMed Material	MAT-10001 — Amoxicillin 500mg Strip
Customer Material No.	APL-AMOX-500
Customer Description	Amoxicillin Caps 500mg x 10
Delivery Priority	02 — High Priority

6.4 Condition Records (Pricing Master)

T-Code: VK11. Condition records store the pricing data for different condition types. AuraMed uses a pricing procedure ZAMSD1 with the following key condition types:

PR00	Base List Price — maintained per material / per customer-material / per customer
K007	Customer-specific discount % — approved by Sales Manager
K020	Price Group discount — bulk buyer tier discount
MWST	Output tax (GST) — 18% on pharmaceutical products above threshold
JCST	GST surcharge — applicable for inter-state sales
FRB1	Freight charge — flat INR 500 or % of order value
SKTO	Cash discount — 2% if paid within 10 days (ZT30 payment terms)

6.5 G/L Accounts (FI Integration)

Revenue account determination in SAP SD is automatic, driven by the Account Assignment Group on the Customer Master and Material Master combined with the Account Key in the Pricing Procedure. Key GL accounts for AuraMed:

800001	Revenue — Domestic Pharmaceutical Sales
800002	Revenue — Domestic Nutraceutical Sales

800010	Revenue — Export Sales (Regulated Markets)
140001	Trade Debtors / Accounts Receivable
500001	Cost of Goods Sold — Finished Goods
220010	GST Payable — CGST Output
220011	GST Payable — SGST Output
220012	GST Payable — IGST Output
430001	Sales Discount Expense

7. End-to-End O2C Process Flow

The Order-to-Cash process for AuraMed consists of the following eight sequential steps. Each step is described with its T-Code, purpose, inputs, outputs, and screen guidance.

Step 1: Customer Inquiry	
T-Code	`VA11 / VA12 / VA13`
Purpose	Captures initial customer interest in buying specific products before a formal quotation is raised. Acts as the starting document for pre-sales activities.
Inputs	Customer number, requested material(s), estimated quantities, required delivery date, sales area (Sales Org / Dist. Channel / Division).
Outputs	Inquiry document (e.g., 10000001). This can be referenced when creating the Quotation to avoid re-entry of data.
Screen	*VA11: 'Create Inquiry' screen opens. Enter Sales Org: AM_S1, Distribution Channel: 10, Division: PH. In the Overview tab, enter Customer: 100001, Material: MAT-10001, Requested Quantity: 500 STR, Requested Delivery Date. Save. System generates Inquiry Number.*

Step 2: Quotation / Sales Quotation	
T-Code	`VA21 / VA22 / VA23`
Purpose	Creates a legally binding offer to the customer detailing prices, quantities, delivery terms, and validity period. The customer receives the quotation and can accept or negotiate. References the Inquiry document.
Inputs	Inquiry reference number (or enter fresh), customer number, material, quantity, requested delivery date, validity start and end dates.
Outputs	Quotation document (e.g., 20000001). Pricing is determined automatically based on condition records. Output (printed/emailed quotation) is triggered via Output Management.
Screen	*VA21: Enter Quotation Type: QT, Sales Org, Dist. Channel, Division. Copy from Inquiry (10000001). In Overview tab, verify pricing — SAP runs pricing procedure ZAMSD1 and shows PR00, K007 discount, GST (MWST). Set Valid From: Today, Valid To: +30 days. Save. System creates Quotation 20000001.*

Step 3: Sales Order

T-Code	**VA01 / VA02 / VA03**
Purpose	The central document of the O2C process. Created after customer confirms the quotation. Triggers credit check, availability check (ATP), and delivery scheduling. This is the binding commitment.
Inputs	Quotation reference number (or fresh entry), customer number, PO number from customer, material(s), confirmed quantities, required delivery date.
Outputs	Sales Order (e.g., 30000001) with confirmed delivery date, net order value, credit status (approved/blocked), scheduled goods issue date.
Screen	*VA01: Order Type: ZOR (Standard Order), Sales Org: AM_S1, Dist. Channel: 10, Div: PH. Enter in Copy From: Quotation 20000001 (system copies all data). Review Header: Pricing tab shows total order value INR 42,500. Conditions tab shows line-item price breakdown. Check Credit Status: Green = approved. Check Availability: Confirmed Qty = 500 STR, Delivery Date = T+3 days. Save. System creates SO 30000001.*

Note: If credit limit is breached, the Sales Order is automatically blocked. A credit block flag appears. T-Code VKM1 (Blocked SD Documents) is used by the Credit Manager to review and release or reject the order.

Step 4: Outbound Delivery Creation	
T-Code	**VL01N / VL02N / VL03N**
Purpose	Creates the physical delivery document against the confirmed Sales Order. Triggers warehouse processes including Transfer Order for picking. The delivery document is the basis for Post Goods Issue (PGI) and billing.
Inputs	Sales Order reference number, shipping point (AM_SP1), delivery date.
Outputs	Outbound Delivery document (e.g., 80000001). Delivery includes picking quantity request, packing instructions, and transport data.
Screen	*VL01N: Enter Shipping Point: AM_SP1, Selection Date (desired delivery date). Enter Order: 30000001. System copies all line items. Review Picking tab — enter Storage Location: SL01 (Finished Goods). Enter Picked Quantity: 500 STR. Perform packing if required. Save Delivery 80000001.*

Step 5: Transfer Order (Picking) — WM	
T-Code	**LT0A / LT1A / LT0I**
Purpose	In Warehouse Management (WM), a Transfer Order instructs the

	warehouse to pick goods from specific bins/storage locations and bring them to the goods issue zone. This step is applicable if AuraMed's plant uses WM (as configured for AM10).
Inputs	Outbound Delivery 80000001, warehouse number (AM_WH), movement type (A — pick for delivery).
Outputs	Transfer Order (TO) number. Picker receives a pick list. On confirmation, the system marks the delivery as 'Picked' and updates bin inventory in WM.
Screen	*LT0A: Enter Warehouse Number: AM_WH, Delivery: 80000001. System proposes source storage type/bin based on storage strategy (FIFO for pharma). Confirm quantities and bin. Save TO. Confirm TO using LT0A / LT12 after physical picking.*

Step 6: Post Goods Issue (PGI)	
T-Code	**`VL02N → Post Goods Issue`**
Purpose	The single most important step in the delivery process. PGI triggers: (1) reduction of inventory stock in MM, (2) automatic posting of Cost of Goods Sold in FI (debit COGS / credit Inventory), and (3) the delivery is now legally 'shipped'. After PGI, billing is possible.
Inputs	Outbound Delivery 80000001 (fully picked and packed), movement type 601 (GI for Sales Order).
Outputs	Goods Issue document (Material Document 4900000001). Inventory reduced by 500 STR of MAT-10001 at Plant AM10, SL01. FI document posted: Dr COGS 500001 / Cr Inventory (GBB). Delivery status changes to 'Goods Issue Posted'.
Screen	*VL02N: Open Delivery 80000001. Click 'Post Goods Issue' button. System immediately posts material document and FI document. Delivery status in header changes from 'A' (Not started) to 'C' (Fully processed). No reversal possible without a return process.*

Note: After PGI, the risk of loss and legal title passes to the customer (per Incoterms EXW). This is the trigger for revenue recognition under Ind AS 115 / IFRS 15.

Step 7: Billing / Invoice Creation	
T-Code	**`VF01 / VF02 / VF04`**
Purpose	Creates the commercial invoice (or proforma) based on the outbound delivery. Triggers automatic FI posting: Dr Accounts Receivable (customer account) / Cr Revenue GL. GST output tax is automatically determined.

	Invoice output (PDF) is generated for dispatch to customer.
Inputs	Outbound Delivery 80000001 (PGI completed), billing date, billing type: F2 (Standard Invoice).
Outputs	Billing Document (e.g., 90000001). FI document posted: Dr Trade Debtors 140001 / Cr Revenue 800001 / Cr GST Payable 220012. Invoice PDF generated and emailed to customer via Output Management.
Screen	*VF01: Enter Billing Type: F2, Billing Date: Today, Delivery: 80000001. System proposes all line items with prices. Review Header: Total: INR 42,500 + GST INR 7,650 = Grand Total INR 50,150. Click Save. Billing Document 90000001 created. Accounting Document 1400000001 posted automatically.*

Step 8: Customer Payment & AR Clearing	
T-Code	**F-28 / F-32 / FB03**
Purpose	Records the incoming payment from the customer against the open invoice. Clears the Accounts Receivable (AR) line item. This is the final step in the O2C cycle — cash has been collected and matched to the invoice.
Inputs	Bank receipt details (amount, value date, bank account), customer account 100001, invoice reference 90000001, payment amount INR 50,150.
Outputs	Cleared FI document. Customer account (140001) balance reduced by INR 50,150. Bank GL account debited. AR item marked as 'Cleared'. DSO (Days Sales Outstanding) KPI updated.
Screen	*F-28: Enter Document Date, Bank Account GL, Amount: 50150. Click 'Process open items'. System lists all open invoices for Customer 100001. Select Invoice 90000001. Click 'Save'. FI clearing document 1500000001 posted. Customer open item cleared.*

7.1 O2C Document Flow Summary

SAP maintains a complete audit trail of the O2C cycle through Document Flow, accessible via T-Code VA03 → Environment → Display Document Flow.

```
  Inquiry: 10000001 (VA11)        └── Quotation: 20000001 (VA21)                └── Sales Order:
30000001 (VA01)                        └── Outbound Delivery: 80000001 (VL01N)
├── Transfer Order: TO-00001 (LT0A)    [WM]                                ├── Goods Issue:
Mat.Doc 4900000001  [MM/FI]                                  └── Billing Document: 90000001
(VF01)                                          ├── FI Invoice: Acc.Doc 1400000001 [FI-AR]
└── Clearing Doc: 1500000001 (F-28) [FI-AR]
```

8. SAP Configuration Steps (SPRO)

All SAP SD configuration is done through SPRO (SAP Project Reference Object). The T-Code for configuration access is SPRO → SAP Reference IMG. The following are the key configuration steps for AuraMed's O2C implementation.

8.1 Enterprise Structure Configuration

Step 1: Define and Assign Sales Organisation

```
SPRO → Enterprise Structure → Definition → Sales & Distribution → Define,
copy, delete, check Sales Organisation
```

- Create Sales Org: AM_S1 (Domestic Sales India) and AM_S2 (Export Sales)
- Assign Sales Organisation to Company Code AM01

Step 2: Define Distribution Channel & Division

```
SPRO → Enterprise Structure → Definition → Sales & Distribution → Define
Distribution Channel
```

- Create: 10-Institutional, 20-Wholesale, 30-Retail, 40-Regulated Export
- Assign Distribution Channel to Sales Organisation AM_S1

Step 3: Set Up Sales Area

```
SPRO → Enterprise Structure → Assignment → Sales & Distribution → Set up
Sales Area
```

- Combine: AM_S1 + 10 + PH, AM_S1 + 20 + PH, AM_S1 + 30 + NU, AM_S2 + 40 + PH

Step 4: Assign Plant to Company Code and Sales Organisation

```
SPRO → Enterprise Structure → Assignment → Logistics – General → Assign
plant to company code
```

- Assign AM10 and AM20 to Company Code AM01

```
SPRO → Enterprise Structure → Assignment → Sales & Distribution → Assign
sales org-distribution channel-plant
```

Step 5: Define and Assign Shipping Points

```
SPRO → Enterprise Structure → Definition → Logistics Execution → Define
Shipping Point
```

- Create Shipping Point AM_SP1 (Hyderabad) and AM_SP2 (Mumbai)
- Assign to Plants AM10 and AM20 respectively

8.2 Sales Document Types Configuration

```
SPRO → Sales & Distribution → Sales → Sales Documents → Sales Document
Header → Define Sales Document Types
```

Sales Document Type: ZOR (Standard Order)

Sales Doc Type	ZOR
Description	AuraMed Standard Sales Order
No. Range (Internal)	03 (from 3000000001 to 3999999999)
Order-Related Billing	Yes (Billing Type: F2)
Delivery Type	LF (Standard Outbound Delivery)
Quotation Messages	QT (reference from Quotation allowed)
Credit Check Active	Yes — with Credit Management (FI-AR)
Output Proc.	V10000 (SAP SD Output Procedure)

8.3 Pricing Procedure Configuration

```
SPRO → Sales & Distribution → Basic Functions → Pricing → Pricing Control →
Define and Assign Pricing Procedures
```

- Create Pricing Procedure: ZAMSD1 (AuraMed Domestic Standard Pricing)
- Configure access sequences for PR00, K007, MWST, FRB1 condition types
- Assign condition types with step numbers, counters, From-To references, and account keys

Pricing Procedure ZAMSD1 — Key Steps

Step	Ctype	Description	From	To	Acct Key
10	PR00	Base Price			
20	K007	Customer Discount %	10		ERL
30	K020	Price Group Discount	10		ERL
40		Net Value after Discount	10	30	
50	MWST	Output Tax (GST)	40		MWS
60	FRB1	Freight Charges			ERF
70		Grand Total	10	60	
80	SKTO	Cash Discount	40		SKT

8.4 Output Determination Configuration

```
SPRO → Sales & Distribution → Basic Functions → Output Control → Output
Determination
```

- Define Output Types: BA00 (Order Confirmation), LD00 (Delivery Note), RD00 (Invoice / Billing Output)
- Assign output condition records: Medium = 5 (External Email), Language = EN
- Partner Function: RE (Bill-to party) receives invoice; WE (Goods recipient) receives delivery note

8.5 Credit Management Configuration

```
SPRO → Financial Accounting → Accounts Receivable & Payable → Credit
Management → Credit Control Account
```

- Define Credit Control Area: AM01
- Assign Credit Control Area to Company Code AM01 (SPRO → FI → AR/AP → Credit Mgmt)
- Set credit limit for each customer via FD32 (transaction, not SPRO)
- Configure credit check at Sales Order and Delivery level
- Set tolerance days and reaction: D = Blocked (error message to sales)

9. Transaction Codes (T-Code) Reference

The following is a comprehensive reference of all SAP transaction codes relevant to the AuraMed O2C implementation:

9.1 Master Data T-Codes

T-Code	Description	Module / Purpose
XD01	Create Customer Master (All Views)	SD / FI-AR
XD02	Change Customer Master	SD / FI-AR
XD03	Display Customer Master	SD / FI-AR
MM01	Create Material Master	MM / SD
MM02	Change Material Master	MM / SD
MM03	Display Material Master	MM / SD
VD51	Create Customer-Material Info Record	SD
VK11	Create Condition Records (Pricing)	SD Pricing
VK12	Change Condition Records	SD Pricing
FD32	Maintain Customer Credit Limit	FI-AR / SD

9.2 Pre-Sales T-Codes

T-Code	Description	Module / Purpose
VA11	Create Customer Inquiry	SD Pre-Sales
VA12	Change Customer Inquiry	SD Pre-Sales
VA13	Display Customer Inquiry	SD Pre-Sales
VA21	Create Quotation	SD Pre-Sales
VA22	Change Quotation	SD Pre-Sales
VA23	Display Quotation	SD Pre-Sales

9.3 Sales Order T-Codes

T-Code	Description	Module / Purpose
VA01	Create Sales Order	SD — Core Process
VA02	Change Sales Order	SD — Core Process
VA03	Display Sales Order / Document Flow	SD — Core Process
VKM1	List Blocked SD Documents (Credit Block)	SD / FI Credit Mgmt
VKM3	Sales Orders Blocked due to Credit	SD / FI Credit Mgmt
VKM4	Delivery Blocked due to Credit	SD / FI Credit Mgmt
V.02	List of Incomplete Sales Orders	SD Monitoring
VA05	List of Sales Orders	SD Reporting

9.4 Delivery & Shipping T-Codes

T-Code	Description	Module / Purpose
VL01N	Create Outbound Delivery (w.r.t. Sales Order)	SD Shipping
VL02N	Change Outbound Delivery / Post Goods Issue	SD Shipping
VL03N	Display Outbound Delivery	SD Shipping
VL06O	Outbound Deliveries for Goods Issue	SD Shipping
LT0A	Create Transfer Order for Delivery (WM)	WM Picking
LT12	Confirm Transfer Order	WM Picking
VT01N	Create Shipment (Transport Planning)	SD Transport
VT02N	Change Shipment	SD Transport

9.5 Billing T-Codes

T-Code	Description	Module / Purpose
VF01	Create Billing Document	SD Billing

T-Code	Description	Module / Purpose
VF02	Change Billing Document	SD Billing
VF03	Display Billing Document	SD Billing
VF04	Process Billing Due List (Mass Billing)	SD Billing
VF11	Cancel Billing Document	SD Billing
MIRO	Logistics Invoice Verification (MM-related)	MM FI
VF21	Create Intercompany Billing	SD Billing

9.6 Finance & AR T-Codes

T-Code	Description	Module / Purpose
F-28	Post Incoming Payment (Manual)	FI-AR
F-32	Clear Customer Open Items	FI-AR
FB03	Display FI Document (Accounting Entry)	FI
FBL5N	Customer Line Item Display (AR Ledger)	FI-AR
F.32	Credit Management — Mass Update	FI Credit Mgmt
VD01	Create Customer (SD View only)	SD
S_ALR_87012178	Customer Balance Confirmation	FI-AR Reporting

9.7 Reporting T-Codes

T-Code	Description	Module / Purpose
VA05	Sales Order List (by Customer / Material)	SD Reporting
VF05	Billing Documents List	SD Reporting
VL06O	Outbound Delivery Monitor	SD Reporting
MB52	Warehouse Stock Overview	MM Inventory
FBL5N	AR Open Item Aging Report	FI-AR Reporting

T-Code	Description	Module / Purpose
`S_ALR_87013019`	Customer Outstanding Dues	FI-AR Reporting
`KKBC_ORD`	Sales Order Profitability	CO Reporting

10. Screen Guidance (Screenshot Descriptions)

Since live SAP system screenshots cannot be included in this report, the following section provides detailed descriptions of each key screen to guide navigation during viva and demonstration.

10.1 VA01 — Create Sales Order Screen

Screen Name	Create Standard Order: Initial Screen
Navigation	SAP Easy Access → Logistics → Sales & Distribution → Sales → Order → VA01
Key Fields	Order Type: ZOR, Sales Organisation: AM_S1, Distribution Channel: 10, Division: PH
Next Screen	Overview tab — enter Sold-to Party (Customer No.), PO Number, Material, Order Qty, Req. Delivery Date
Important Tabs	Header: Pricing, Partners, Shipping, Billing | Item: Conditions, Schedule Lines, Shipping
System Response	SAP runs: (1) Partner determination, (2) Pricing procedure ZAMSD1, (3) ATP availability check, (4) Credit check. All results shown with traffic light indicators (green/yellow/red).

10.2 VL01N — Create Outbound Delivery Screen

Screen Name	Create Outbound Delivery: Initial Screen
Navigation	SAP Easy Access → Logistics → Sales & Distribution → Shipping & Transportation → Outbound Delivery → Create → Single Document → With Reference to Sales Order → VL01N
Key Fields	Shipping Point: AM_SP1, Selection Date (Delivery Date), Order: 30000001
Overview Screen	Shows all line items with delivery quantity. Click 'Picking' tab to enter Storage Location SL01 and Picked Quantity.
PGI Button	'Post Goods Issue' button appears in the toolbar after picking confirmation. Click to trigger stock reduction and FI posting.
Status	Header status changes: Overall: A→C, Good Movement Status: blank→C after PGI.

10.3 VF01 — Create Billing Document Screen

Screen Name	Create Billing Document: Initial Screen
Key Fields	Billing Type: F2 (Invoice), Billing Date: current date, enter Delivery no. 80000001
Overview Screen	Billing document header shows: Customer, Billing Date, Net Amount, Tax Amount, Total Amount. Line items show individual product prices.
Accounting Tab	After save, click Accounting tab to view the FI document number posted. Shows Dr Debtors / Cr Revenue / Cr Tax.
Output	Output type RD00 triggers PDF invoice generation. Email can be sent directly to customer's accounts payable team.

10.4 F-28 — Post Incoming Payment Screen

Screen Name	Post Incoming Payments: Header Data Screen
Key Fields	Document Date, Company Code: AM01, Currency: INR, Bank Account GL, Amount, Value Date
Process Items	Click 'Process Open Items'. System shows all unpaid invoices for the customer. Select Invoice 90000001. Difference = zero = fully matched.
Save	FI clearing document created. Customer account updated. Invoice moves from open items to cleared items (visible in FBL5N with status 'Cleared').

11. Integration with Other SAP Modules

11.1 SD — FI Integration

SAP SD is tightly integrated with Financial Accounting (FI) at two critical junctures:

At Post Goods Issue (PGI) — SD–MM–FI

- Movement Type 601 (Goods Issue for Sales Order) is triggered automatically
- FI Posting: Dr Cost of Goods Sold (GL 500001) / Cr Finished Goods Inventory (GL 310001)
- This posting is driven by the Account Determination settings in OBYC (T-Code) for transaction key GBB / VAX

At Billing (Invoice) — SD–FI

- FI Posting: Dr Trade Debtors / Customer Account (GL 140001) / Cr Revenue GL (800001) / Cr GST Payable (220012)
- Revenue account is determined by Account Assignment Group on Customer + Material Master combined with Account Key in Pricing Procedure
- Billing document and FI accounting document have a 1:1 relationship — linked via accounting document number

11.2 SD — MM Integration

The O2C process involves MM at the Delivery and PGI stage:

- ATP (Available-to-Promise) check during Sales Order entry queries MM stock tables (MARD, MARA) in real time
- Goods Issue (PGI) reduces stock using MM movement type 601 — inventory reduced at Plant AM10, SL01
- Material valuation (Standard Price from Material Master Accounting view) determines the value of goods issued for COGS posting
- Returns processing: Movement type 651 (returns delivery) and 655 (put back to stock after quality check) are triggered from SD return orders

11.3 SD — PP Integration

When pharmaceutical products are produced to order (Make-to-Order scenario):

- Sales Order triggers a Production Order in PP via MRP run (MD01 / MD02)
- Production Order is linked to the Sales Order line item — stock produced is reserved exclusively for that sales order
- Production completion (CO15 Confirmation) updates MM stock which is then available for delivery against the Sales Order

- Schedule line in Sales Order is updated with production confirmation dates from PP

11.4 Integration Overview Diagram

```
  ┌──────────────────────────────────────────────────────┐   │                    SAP SD
(Sales & Distribution)      │   │  Inquiry → Quotation → Sales Order → Delivery →      │   │
PGI → Billing → Payment Clearing                      │
└──────┬──────────────────────────┴──────────────────┴──────┘          │                       │
│          ▼                        ▼                ▼   ┌──────────────┐
┌──────────────────┐  ┌──────────────┐   │   SAP FI     │    │      SAP MM         │  │
SAP PP     │   │ — AR Posting│    │ — ATP Stock Check  │  │  — Prod Order│   │ — Rev. Rec.
│    │  — Goods Issue 601  │  │  — MTO Stock │   │ — GST Tax    │    │  — Inventory Mgmt   │
│  — MRP Run    │   │ — AR Clear  │    │ — Returns Mgmt     │  │  — Confm.    │
└──────────────┘   └──────────────────┘  └──────────────┘  └──────────────┘
```

12. ABAP Enhancements (User Exits & BADIs)

AuraMed required several custom business logic extensions to the standard SAP SD process. These were implemented using SAP-standard enhancement techniques: User Exits and BADIs (Business Add-Ins). Custom Z-programs were developed for reporting.

12.1 User Exit: MV45AFZZ — Sales Order Custom Validation

Enhancement Spot	SAPMV45A (Sales Order Processing)
Form Routine	USEREXIT_SAVE_DOCUMENT_PREPARE (in MV45AFZZ)
Business Need	AuraMed requires that all export Sales Orders (Sales Org = AM_S2) must have a customer PO number entered. Standard SAP does not enforce this.
Logic Implemented	Custom ABAP code checks: IF vbak-vkorg = 'AM_S2' AND vbak-bstnk IS INITIAL. IF true, MESSAGE E999(ZSD) 'Export Orders must have Customer PO Reference. Please enter in field PO Number.' ENDIF
Result	Order cannot be saved without Customer PO Number for export sales. Reduces disputes and customs documentation errors.

Code Snippet — USEREXIT_SAVE_DOCUMENT_PREPARE

```
FORM userexit_save_document_prepare.
  " Check: Export orders must have Customer PO number
  IF vbak-vkorg = 'AM_S2'
    AND vbak-bstnk IS INITIAL.
    MESSAGE e999(zsd) WITH
      'Export Orders require Customer PO Reference'.
  ENDIF.
ENDFORM.
```

12.2 BAdI: BADI_SD_SALES_ITEM — Minimum Order Quantity Check

BAdI Name	BADI_SD_SALES_ITEM
Method Used	CHANGE_AT_SAVE
Business Need	For wholesale distributor channel (Distribution Channel 20), AuraMed enforces a minimum order quantity of 100 boxes per line item to ensure economical order processing.

Logic Implemented	Custom implementation class ZCL_SD_SALES_ITEM_IMPL. In method CHANGE_AT_SAVE: Loop through all order items. If distribution channel = '20' and order quantity < 100 BOX: raise error message.
Result	Prevents sub-minimum orders from wholesalers, reducing order processing cost per line.

12.3 User Exit: MV50AFZ1 — Batch Number Mandatory at Delivery

Enhancement	SAPMV50A — Delivery Processing User Exit MV50AFZ1
Form Routine	USEREXIT_SAVE_DOCUMENT_PREPARE
Business Need	Pharmaceutical regulations (Schedule M, GMP) require batch number traceability for all dispatched medicines. SAP does not mandate batch number entry by default.
Logic	Check all delivery line items: IF material type = 'FERT' (finished pharma goods) AND batch field (lips-charg) IS INITIAL: raise error 'Batch Number is mandatory for pharmaceutical products'.
Result	100% batch traceability ensured for regulatory compliance. Eliminates risk of dispatching products without batch reference.

12.4 Custom ABAP Report: ZSD_O2C_STATUS — O2C Status Tracker

Program Name	ZSD_O2C_STATUS
Transaction	ZO2C (custom Z-transaction created via SE93)
Business Need	Sales managers needed a single-screen view of all orders showing current O2C status: order confirmed / delivery created / PGI done / billed / payment received.
Selection Screen	Customer Number, Sales Organisation, Date Range (Order Creation Date), Material
Output ALV Grid	Sales Order No. | Customer | Material | Order Qty | Delivery No. | PGI Date | Invoice No. | Invoice Amount | Payment Status | Outstanding Amount
Key ABAP Objects	Tables: VBAK (Sales Order Header), VBAP (Item), LIKP (Delivery Header), LIPS (Delivery Item), VBRK (Billing Header), BSID (Open AR Items). Joined via VBFA (Document Flow) table.

12.5 Enhancement Framework Summary

The following table summarises all custom developments implemented for AuraMed:

Enhancement / BADI	Program / Class	Type	Business Purpose
MV45AFZZ	User Exit Form	User Exit	Export Order PO Number validation
BADI_SD_SALES_ITEM	ZCL_SD_SALES_ITEM_IMPL	BAdI	Min. order qty for wholesale channel
MV50AFZ1	User Exit Form	User Exit	Batch number mandatory at delivery
ZSD_O2C_STATUS	ABAP Report Program	Custom Report	O2C order status tracker dashboard

13. Challenges & Solutions

Challenge 1: Master Data Migration from Legacy Systems

Challenge	AuraMed had customer and material master data spread across 4 legacy systems with inconsistent naming conventions, duplicate records, and missing GST/tax data. Migrating ~8,000 customer records and ~3,500 material records was complex.
SAP Solution	SAP LSMW (Legacy System Migration Workbench) and SAP Data Services were used. A data cleansing exercise was conducted prior to migration. Data mapping templates were created for Customer Master (XD01) and Material Master (MM01) upload. Three rounds of data validation were conducted using custom ABAP programs before final migration.

Challenge 2: Complex Pharmaceutical Pricing Tiers

Challenge	AuraMed's pricing was highly complex — different prices for government tenders, private hospitals, wholesalers, and retail. Over 12,000 customer-material-specific price records needed to be loaded.
SAP Solution	SAP's Condition Technique in SD Pricing was configured with multiple access sequences. A condition table with combination (Customer / Material / Sales Org) was created. Batch upload of condition records was done using custom ABAP program ZVK11_UPLOAD reading from Excel/CSV files into condition tables (A-tables in KONP/KONV).

Challenge 3: GST E-Invoicing Integration

Challenge	India's GST regulations require businesses with turnover above INR 5 Crore to generate e-invoices through the government's Invoice Registration Portal (IRP). AuraMed's SAP billing had to integrate with GSTN's IRP in real time.
SAP Solution	SAP's Advanced Compliance Reporting (ACR) and the GST plug-in for SAP were implemented. Custom output type ZIRN (IRN Generation) was added to the billing output procedure. On billing save, an RFC call is triggered to the IRP API. IRP returns an IRN (Invoice Reference Number) and QR code

	which are printed on the invoice. Implemented via BAdI: FI_TAX_BADI_INDIR_TAX.

Challenge 4: User Resistance and Change Management

Challenge	AuraMed's field sales team (250+ users) was accustomed to receiving orders via WhatsApp and phone. Transitioning to SAP order entry was met with significant resistance. The warehouse team was unfamiliar with system-driven picking.
SAP Solution	A comprehensive training programme was designed using SAP Enable Now (WalkMe-style training tool). Role-based training was delivered: Sales Team (VA01, VA03), Warehouse (VL01N, LT0A), Finance (VF01, F-28, FBL5N). A 4-week parallel run was conducted where both legacy and SAP systems ran simultaneously. Hypercare support was provided for 8 weeks post go-live.

Challenge 5: Credit Block Dispute Resolution Process

Challenge	When SAP's credit management automatically blocked orders, there was no defined business process for escalation and release. Sales managers were overriding credit blocks without proper justification, defeating the purpose of the system.
SAP Solution	A Credit Release Workflow was implemented using SAP Business Workflow (via T-Code SWDD). When an order is blocked, an automated email is sent to the Credit Manager with order details. The Credit Manager reviews via VKM1/VKM3 and either releases (with mandatory reason code entry) or rejects. The Sales Representative receives an automated notification of the decision. All release decisions are logged in a custom Z-table for audit purposes.

14. Conclusion

14.1 Benefits Realised After SAP O2C Implementation

The SAP SD-based O2C implementation at AuraMed Pharmaceuticals delivered the following measurable business benefits within the first year of go-live:

KPI / Metric	Pre-SAP (As-Is)	Post-SAP (To-Be)
Order Processing Time	4–6 hours	< 30 minutes
Invoice Error Rate	8.5%	< 0.5%
Days Sales Outstanding (DSO)	62 days	41 days
Credit Note Volume (monthly)	~180 credit notes	~12 credit notes
Stock-out at order entry	22% of orders	< 3% of orders
Revenue Report Latency	15–20 days	Real-time (SAP Fiori)
Bad Debt Write-off (annual)	INR 45 Crore	INR 8 Crore
GST Compliance Violations	12 notices in FY23	0 notices in FY25

14.2 Learning Outcomes

Through the design, configuration, and documentation of this SAP O2C project, the following key learning outcomes were achieved:

- Gained deep understanding of the Order-to-Cash business cycle and its mapping to SAP SD standard processes.
- Learnt the SAP SD organisational structure including Sales Organisation, Distribution Channel, Division, and how they combine into a Sales Area.
- Understood the SAP Condition Technique for pricing — how access sequences, condition tables, condition types, and pricing procedures work together.
- Developed skills in SAP Master Data management: Customer Master (XD01), Material Master (MM01), Customer-Material Info Records (VD51), and Pricing Condition Records (VK11).
- Acquired knowledge of SAP MM–SD integration through Availability Check (ATP) and Post Goods Issue (movement type 601).
- Understood SAP FI–SD integration: how billing triggers automatic revenue and tax postings, and how payment clearing closes the O2C cycle.
- Applied ABAP enhancement skills: User Exits (MV45AFZZ, MV50AFZ1) and BAdIs (BADI_SD_SALES_ITEM) for custom business logic.
- Learnt real-world implementation challenges such as data migration, change management, GST e-invoicing integration, and credit workflow design.

14.3 Future Scope

- Implementation of SAP Customer Experience (CX / C4C) integration with SD for a complete front-office to back-office O2C digital thread.
- Adoption of SAP S/4HANA Intelligent Order Management for AI-driven order promising and dynamic route optimisation.
- Integration with India's Account Aggregator framework for automated bank statement import and faster AR reconciliation.
- Extension of the O2C process to include subscription-based billing for AuraMed's hospital AMC (Annual Maintenance Contract) customers.

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— End of Report —

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